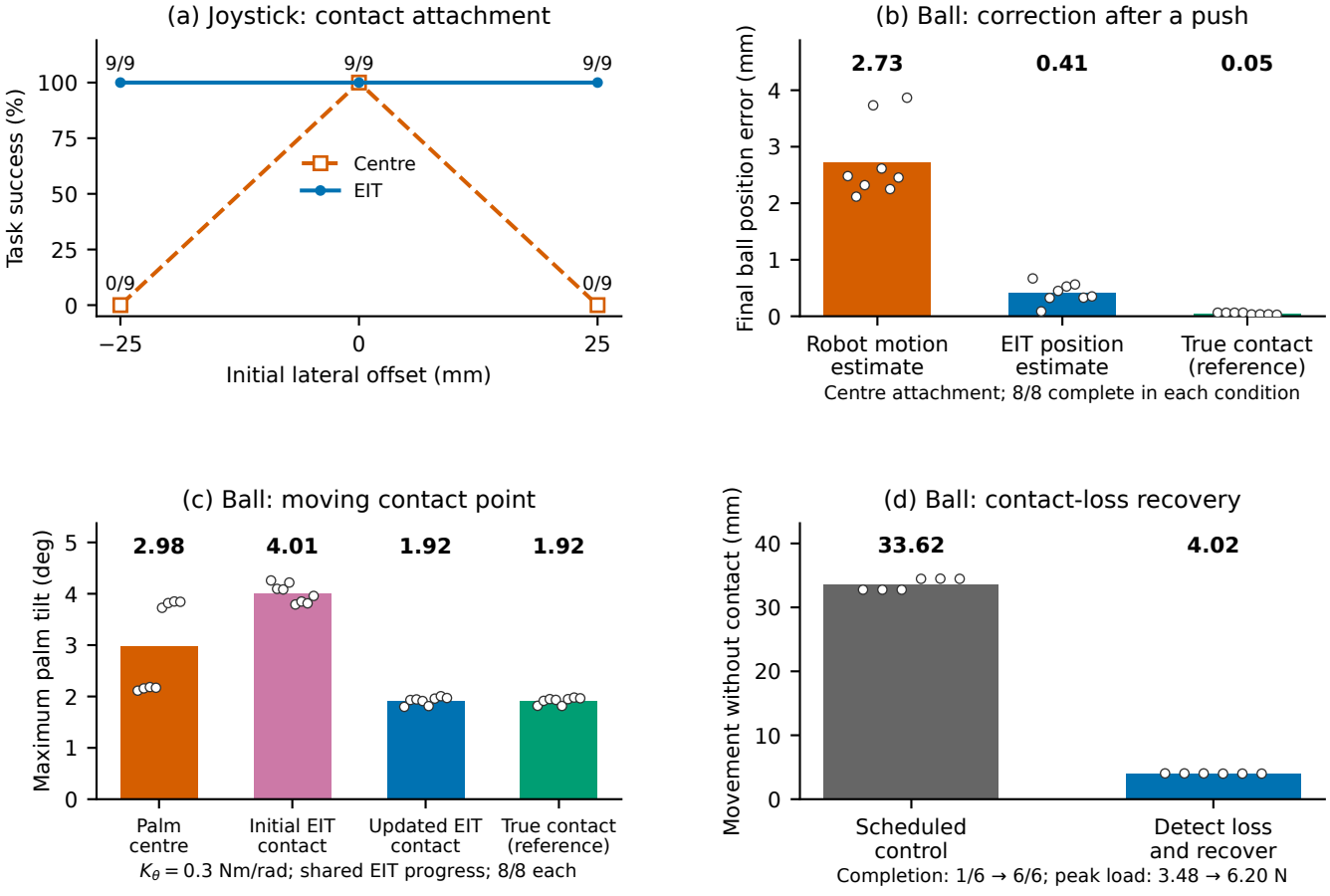
(a) Joystick: contact attachment
Task success (%)
100
75
50
25
0
9/9
9/9
9/9
0/9
0/9
Centre
EIT
−25
0
25
Initial lateral offset (mm)
(b) Ball: correction after a push
Final ball position error (mm)
2.73
0.41
0.05
Robot motion estimate
EIT position estimate
True contact (reference)
Centre attachment; 8/8 complete in each condition
(c) Ball: moving contact point
Maximum palm tilt (deg)
2.98
4.01
1.92
1.92
Palm centre
Initial EIT contact
Updated EIT contact
True contact (reference)
$K_\theta = 0.3$ Nm/rad; shared EIT progress; 8/8 each
(d) Ball: contact-loss recovery
Movement without contact (mm)
33.62
4.02
Scheduled control
Detect loss and recover
Completion: 1/6 → 6/6; peak load: 3.48 → 6.20 N